Haven Project Overview

Citizen Safety and Complaint Rescue Platform

Haven is a university project for Pakistan that gives citizens one clear place to report personal safety and civic problems. It helps a person explain what happened, receive short next-step guidance, and follow the case while authorized people review and act on it.

Core principle AI suggests and humans decide

Project in brief

Haven covers two tracks. The Safety and Justice track gives higher attention to harassment, stalking, threats, domestic violence, theft, blackmail, fraud, missing-person concerns, and other personal safety cases. The Civic track handles issues such as damaged roads, drainage, garbage, unsafe electricity, water problems, and street lights.

The platform does not guarantee police or rescue attendance. Its purpose is to reduce confusion before and after a report: guide the person toward safer immediate actions, organise the complaint, send it to the correct human reviewer, and make progress visible.

Why Haven is needed

People may delay asking for help because they are frightened, unsure which department to contact, uncomfortable using formal language, or worried that nobody will take the complaint seriously. Haven lowers that first barrier. A person can begin in familiar words and review the prepared summary before submitting it.

Feminist design focus

Haven gives particular attention to women and girls who may face street harassment, stalking, domestic violence, workplace harassment, online abuse, threats, or blackmail. These situations often involve fear, social pressure, financial dependence, or an imbalance of power. A useful system must therefore protect agency and privacy rather than forcing a survivor through a confusing process.

- A woman can describe the incident in English, Urdu, or Sindhi without knowing legal terminology.
- She can review or correct the summary so the system does not take control of her story.
- Public anonymity and restricted staff access reduce unnecessary exposure of identity.
- Danger signals such as threats, stalking, weapons, or an attack raise urgency and require human review.
- Safety guidance prioritizes moving to safety and contacting trusted people or emergency services when needed.
- An evidence checklist helps preserve messages, dates, locations, photos, or recordings only when collection is safe.
- Status updates reduce the need to repeatedly retell a distressing experience to different offices.

This approach reflects a feminist concern with access, voice, consent, privacy, and accountability. Haven supports the person's choices while keeping final operational decisions with authorized human reviewers.

Who benefits

The main users are women and other citizens facing safety problems, people reporting local civic hazards, families helping a relative, and people who communicate more comfortably in Urdu or Sindhi. Staff also benefit because complaints arrive in a consistent format with urgency signals, evidence, location details, and a recorded history of every decision.

Main features

Feature	How it helps
Accessible reporting	A citizen can start with text and attach image or audio evidence. The experience supports English, Urdu, and Sindhi, including right-to-left layouts.
Safety first analysis	Transparent keyword rules check for immediate danger before AI runs. Weapons, attacks, severe medical danger, fire, stalking, threats, and serious civic hazards can raise urgency.
Optional Gemini support	Gemini can prepare a short summary, translation, category, urgency suggestion, and practical next steps. Complaint text is treated as untrusted data.
Safe AI fallback	If Gemini is unavailable or produces invalid output, Haven keeps working, raises the case to at least High, creates local guidance, and requires human review.
Human control room	Authorized staff review the original statement, AI summary, rule matches, evidence, location, and timeline before accepting, requesting information, rejecting, or assigning a team.
Live case tracking	Citizens see a case number, timeline, assignment, estimated arrival when provided, and response updates such as On the way, Arrived, and Closed.
Private evidence and documents	Evidence stays in private storage and opens through short-lived links. The API can create complaint applications, evidence checklists, civic applications, and factual FIR drafts.
Appeal and reopen options	A citizen can appeal a decision or reopen an eligible rejected or closed case with a reason. These actions remain part of the audit history.
Security and accountability	Role-based access, Row Level Security, rate limits, validation, duplicate checks, and audit logs protect the demo and record sensitive actions.
Installable web apps	Separate citizen and staff applications work on desktop and mobile and can be installed as progressive web apps.

How a case moves through Haven

The workflow separates automated assistance from human authority. Each step gives the citizen a clear result and creates a record for the next responsible person.

Step	Action	Result
1	Start the report	The citizen signs in, selects a safety or civic category, and explains the problem in their own words.
2	Add details	The citizen adds location, privacy choice, and available evidence. An unfinished report can remain a draft.
3	Run safety rules	English, Urdu, and Sindhi danger phrases are checked first. A matched danger signal can only raise urgency.
4	Prepare the case	Gemini may summarize, translate, suggest a category, and suggest urgency. The citizen's original statement is preserved.
5	Set safe priority	Haven uses the higher result from the safety rules and AI. AI cannot lower the rule-based urgency.
6	Give immediate guidance	The citizen receives short safety steps, practical next actions, useful information, and an evidence checklist.
7	Human review	A control-room admin or department officer checks the case and accepts it, requests more information, rejects it with a reason, or assigns an appropriate team.
8	Response and tracking	An assigned response team records On the way, Arrived, and Closed updates. Citizens receive in-app notifications and can follow the timeline.
9	Follow-up	The citizen can download prepared documents, file an appeal, or reopen an eligible case. Every important action is audit-logged.

Users and responsibilities

Role	Responsibility
Citizen	Creates and tracks personal complaints, receives guidance and notifications, accesses private evidence and documents, and may appeal or reopen eligible cases.
Control Room Admin	Reviews Safety and Justice cases, confirms urgency, records decisions, and assigns a suitable response or referral team.
Department Officer	Reviews Civic cases and routes accepted complaints to the correct department or civic team.
Response Team	Sees only actively assigned cases and records operational progress from assignment to closure.
Super Admin	Has cross-track oversight and can review users, audit information, safety playbooks, and legal-content verification states.

Support alongside existing services

Haven is designed to work alongside police, rescue services, legal aid, women's rights organisations, community groups, and local government departments. It does not replace them. A report can make the next step clearer, but it cannot guarantee that police or a team will arrive, and it must not show an arrival promise that was not provided by the responsible dispatcher.

- Immediate danger still requires direct contact with the appropriate emergency service.
- AI does not dispatch teams, decide guilt, give a legal verdict, or reject a complaint.
- Legal information is general guidance and must be verified by qualified people before real-world publication.
- Generated FIR material is a factual draft for officer confirmation, not an official FIR.
- The current university version uses demo teams and in-app updates rather than live authority integration.

Technology and privacy

The system uses React and TypeScript for the citizen and staff applications, Node.js and Express for the protected API, Supabase for authentication, PostgreSQL data, private storage, and real-time updates, and optional Google Gemini assistance. Citizens see only their own cases. Staff access is limited by role and track, while response teams see only assigned cases. Signed links protect evidence and generated files, and audit records show who performed each sensitive action.

Overall project value

Haven demonstrates how technology can make complaint reporting more accessible without handing authority to AI. Its strongest contribution is the combination of survivor-centred reporting, multilingual access, transparent safety rules, human accountability, and visible follow-up. For women facing harassment or violence, this can provide a clearer and more private first step. For other citizens, it provides the same structured path for safety and civic concerns.